Water Quality Monitoring Report

Lake Liddell, Hunter Valley NSW

Analysis period: 2021-01-14 to 2024-12-29 | Generated: 2026-07-10

1. Study Context

Lake Liddell is located in the Hunter Valley, NSW, adjacent to the former Liddell coal-fired power station, which ceased operations in April 2022. The lake historically served as a cooling water reservoir and is subject to water quality monitoring obligations under NSW environmental legislation. This report presents an automated remote sensing analysis of water quality proxies derived from Sentinel-2 L2A imagery acquired via Microsoft Planetary Computer.

2. Methodology

Sentinel-2 L2A scenes with less than 20% cloud cover were retrieved via the Planetary Computer STAC API for the period January 2021 to December 2024. Scenes were filtered to the Lake Liddell AOI polygon, cloud-masked using the Scene Classification Layer (SCL), and three water quality indices computed: NDWI (McFeeters 1996), NDTI (Lacaux 2007), and a turbidity proxy (B04/B03 ratio). Anomaly detection used a post-transition baseline (April 2022 onwards) with scenes exceeding 2 standard deviations flagged for review.

3. Dataset Summary

Total scenes retrieved	267
Scenes after cloud filtering	259
Scenes after deduplication	119
Scenes used in analysis	119
Analysis period	2021-01-14 to 2024-12-29
Cloud cover threshold	<20%
Minimum valid water pixels	2,000

4. Post-Transition Baseline Statistics (Apr 2022 - Dec 2024)

Index	Mean	Std Dev	Scenes
NDWI	0.0844	0.0210	86
NDTI	-0.0575	0.0135	86
Turb B04/B03	0.8917	0.0244	86

5. Anomalous Scenes Detected

7 scenes exceeded the 2-standard-deviation threshold on at least one index. Early post-closure scenes (April-May 2022) show elevated turbidity consistent with sediment resuspension following cessation of active water management. Scenes in late 2023 and 2024 show inverse anomalies (elevated NDWI, reduced turbidity) consistent with above-average rainfall and increased water clarity in the Hunter Valley catchment.

Date	NDWI	NDTI	Turb	NDWI z	NDTI z	Turb z
2022-04-04	0.0467	-0.0275	0.9465	-1.80	+2.22	+2.25
2022-05-14	0.0325	-0.0104	0.9796	-2.48	+3.48	+3.60

2023-03-15	0.0545	-0.0289	0.9446	-1.43	+2.11	+2.17
2023-04-09	0.0373	-0.0285	0.9451	-2.25	+2.14	+2.19
2023-12-30	0.1145	-0.0853	0.8429	+1.44	-2.06	-2.00
2024-02-03	0.1292	-0.0771	0.8570	+2.14	-1.45	-1.42
2024-11-09	0.1285	-0.0963	0.8244	+2.10	-2.87	-2.76

6. Figures

Figure 1: Water quality index time series 2021-2024

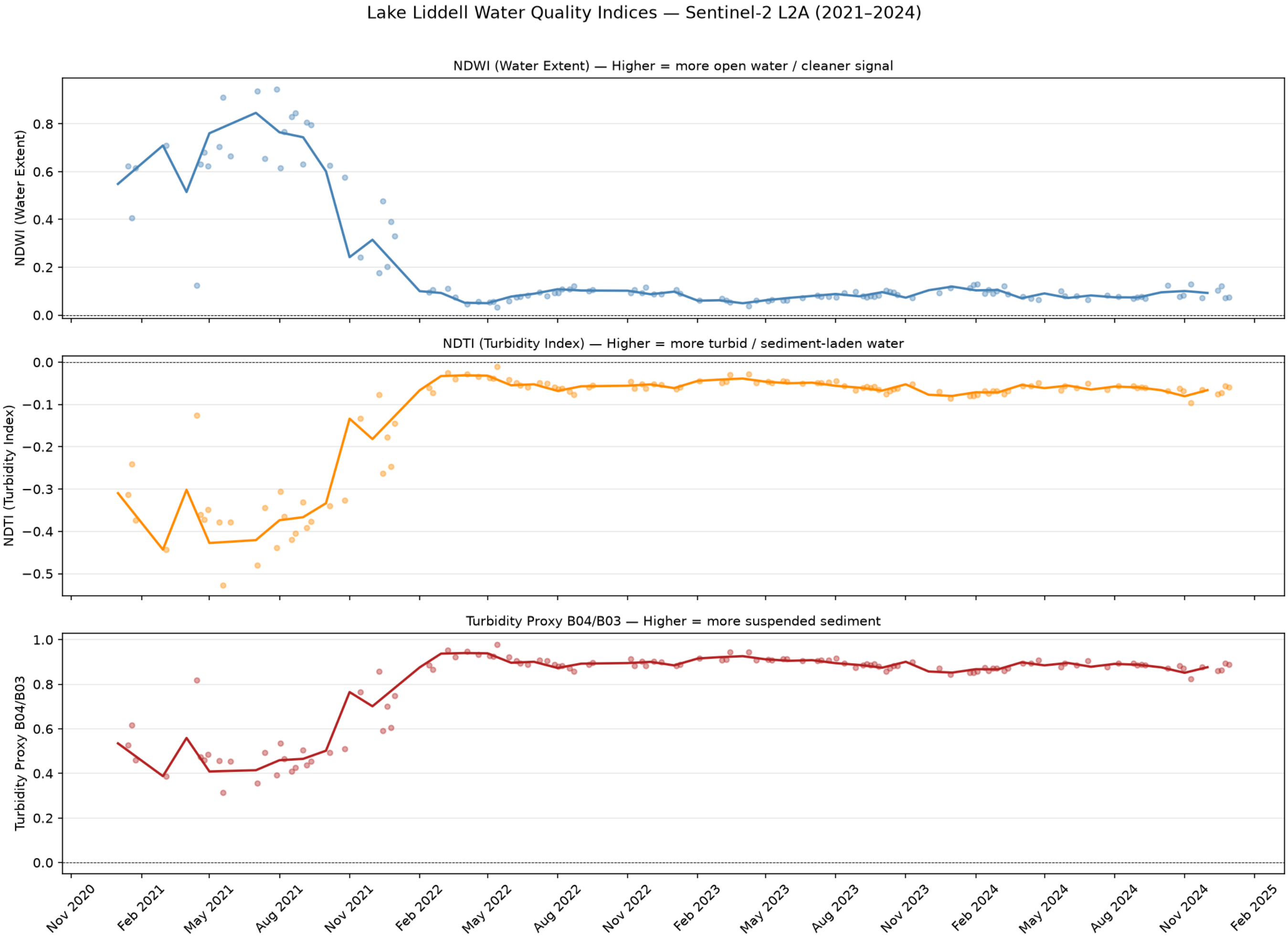

Figure 2: Post-transition anomaly detection 2022-2024

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Lake Liddell — Post-Transition Water Quality Anomalies (2022–2024)